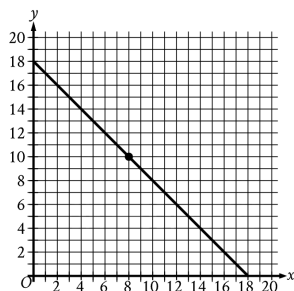


## Module-1

1.



The graph in the  $xy$ -plane models the possible combinations of length  $x$ , in meters (m), and width  $y$ , in meters, for a rectangle with a perimeter of 36 m. Which statement is the best interpretation of the point  $(8, 10)$  in this context?

- ☐ A The length is 10 m less than the perimeter, and the width is 8 m less than the perimeter.
- ☐ B The length is 10 m, and the width is 8 m.
- ☐ C The length is 8 m, and the width is 10 m.
- ☐ D The length is 8 m less than the perimeter, and the width is 10 m less than the perimeter.

2.

$$24.5x + 24.75y = 641$$

Isabel ordered topsoil and crushed stone, which cost a total of \$641, for her garden. The given equation represents the relationship between the number of cubic yards of topsoil,  $x$ , and the number of tons of crushed stone,  $y$ , Isabel ordered. How much more, in dollars, did a ton of crushed stone cost Isabel than a cubic yard of topsoil?

3.

A chemist studying the impact of salt on a process mixes  $x$  kilograms of a low-salt mixture, which is 2% salt by weight, with  $y$  kilograms of a high-salt mixture, which is 96% salt by weight, to create 24 kilograms of a mixture that is 4% salt by weight. Which equation represents this situation?

- ☐ A  $0.96x + 0.02y = (0.04)(24)$
- ☐ B  $0.02x + 0.96y = (0.04)(24)$
- ☐ C  $0.96x + 0.02y = 24$
- ☐ D  $0.02x + 0.96y = 24$

4.

A company that creates and sells tape dispensers calculates its monthly profit, in dollars, by subtracting its fixed monthly costs, in dollars, from its monthly sales revenue, in dollars. The equation  $15,000 = 2.00x - 4,500$  represents this situation for a month where  $x$  tape dispensers are created and sold. Which statement is the best interpretation of  $2.00x$  in this context?

- ☐ A The monthly sales revenue, in dollars, from selling  $x$  tape dispensers
- ☐ B The monthly sales revenue, in dollars, from each tape dispenser sold
- ☐ C The monthly cost, in dollars, of creating each tape dispenser
- ☐ D The monthly cost, in dollars, of creating  $x$  tape dispensers

5.

The equation  $7g + 7b = 840$  represents the number of blue tiles,  $b$ , and the number of green tiles,  $g$ , an artist needs for an 840-square-inch tile project. The artist needs 71 blue tiles for the project. How many green tiles does he need?

6.

| Distance (kilometers) | Average time (minutes) |
|-----------------------|------------------------|
| 0.32                  | 8                      |
| 0.56                  | 14                     |
| 0.68                  | 17                     |

The table gives the average time  $t$ , in minutes, it takes Carly to travel a certain distance  $d$ , in kilometers. Which equation could represent this linear relationship?

☐ A  $t = 4d$

☐ B  $t = \frac{1}{25}d$

☐ C  $t = 25d$

☐ D  $t = \frac{1}{4}d$

7.

Lily made 36 cups of jam. Lily then filled  $x$  small containers and  $y$  large containers with all the jam she made. The equation  $4x + 6y = 36$  represents this situation. Which is the best interpretation of  $6y$  in this context?

☐ A The number of large containers Lily filled

☐ B The number of small containers Lily filled

☐ C The total number of cups of jam in the large containers

☐ D The total number of cups of jam in the small containers

8.

A batch of banana milkshakes consists of 4 cups of ice cream and 2 bananas and has 1,114 milligrams (mg) of calcium. There is 276 mg of calcium in 1 cup of the ice cream used to make this batch of milkshakes. How much calcium, in mg, is in 1 banana?

☐ A 5

☐ B 10

☐ C 419

☐ D 1,104

9.

In science class, Diego conducted an experiment to learn about evaporation. Diego measured the height of fluid in a beaker over a period of time. The function  $f(x) = 39 - 0.18x$  gives the estimated height, in centimeters (cm), of the fluid in the beaker  $x$  days after the start of the experiment. Which of the following is the best interpretation of 39 in this context?

☐ A The estimated height, in cm, of the fluid at the start of the experiment

☐ B The estimated height, in cm, of the fluid at the end of the experiment

☐ C The estimated change in the height, in cm, of the fluid each day

☐ D The estimated number of days for all the fluid to evaporate

10.

A museum rents tablets to visitors. The museum earns revenue of \$14 for each tablet rented for the day. On Wednesday, the museum earned \$406 in profit from renting tablets after paying daily expenses of \$112. How many tablets did the museum rent on Wednesday? (profit = total revenue – total expenses)

11.

The sum of a number  $x$  and 7 is twice as large as a number  $y$ . The number  $y$  is 3 less than the number  $x$ . Which system of equations describes this situation?

☐ A  $x + 7 = 2y$   
 $y = x - 3$

☐ B  $x + 7 = 2y$   
 $y = 3 - x$

☐ C  $2(x + 7) = y$   
 $y = x - 3$

☐ D  $2(x + 7) = y$   
 $y = 3 - x$

12.

A particular botanist classifies a species of plant as tall if its typical height when fully grown is more than 100 centimeters. Each of the following inequalities represents the possible heights  $h$ , in centimeters, for a specific plant species when fully grown. Which inequality represents the possible heights  $h$ , in centimeters, for a tall plant species?

☐ A  $106 < h < 158$

☐ B  $80 < h < 100$

☐ C  $42 < h < 87$

☐ D  $17 < h < 85$

13.

A city employee will plant two types of bushes, azaleas and boxwoods, in a park. There will be no more than 164 total bushes planted, and the number of azaleas planted will be at most three times the number of boxwoods planted. Which of the following systems of inequalities best represents this situation, where  $a$  is the number of azaleas that will be planted, and  $b$  is the number of boxwoods that will be planted?

- ☐ A  $a + b \geq 164$   
 $3a \geq b$
- ☐ B  $a + b \geq 164$   
 $a \leq 3b$
- ☐ C  $a + b \leq 164$   
 $3a \geq b$
- ☐ D  $a + b \leq 164$   
 $a \leq 3b$

14.

The cost to rent a commercial fishing boat from a certain company is \$950 for the first 2 hours and an additional \$50 per hour for each hour after the first 2 hours. If the total cost to rent the commercial fishing boat from the company for  $t$  hours, where  $t > 2$ , is \$1,100, which equation represents this situation?

- ☐ A  $950(t - 2) + 50t = 1,100$
- ☐ B  $950(2t) + 50t = 1,100$
- ☐ C  $950 + 50(t - 2) = 1,100$
- ☐ D  $950 + 50(2t) = 1,100$

15.

A geologist estimates that the volume of a slab of granite is greater than 12.7 cubic feet but less than 15.7 cubic feet. The geologist also estimates that the slab of granite weighs 165 pounds per cubic foot of volume. Which inequality represents this situation, where  $x$  represents the estimated total weight, in pounds, of the slab of granite?

- ☐ A  $165 - 15.7 < x < 165 - 12.7$
- ☐ B  $165 + 12.7 < x < 165 + 15.7$
- ☐ C  $165(12.7) < x < 165(15.7)$
- ☐ D  $\frac{165}{15.7} < x < \frac{165}{12.7}$

16.

| Area (square feet) | Water (gallons) |
|--------------------|-----------------|
| 2,520              | 4,536           |
| 3,780              | 6,804           |
| 5,040              | 9,072           |

The buildings of a shopping center are designed to allow water to drain from the roof into gutters on the sides of the buildings. The table shows the relationship between the area  $x$ , in square feet, of a roof and the amount of water  $f(x)$ , in gallons, drained from the roof into the gutters over a certain period of time. Which equation could define  $f$ ?

- ☐ A  $f(x) = 0.6x$
- ☐ B  $f(x) = 1.8x$
- ☐ C  $f(x) = 2,268x$
- ☐ D  $f(x) = 4,536x$

17.

In 2010, a swim club had a total of 35 swimmers, each classified as either advanced or intermediate. From 2010 to 2020, the number of advanced swimmers in the club increased by approximately 53%, and the number of intermediate swimmers in the club increased by approximately 44%. The total number of swimmers in the club increased by approximately 49%. Which equation best represents this situation, where  $a$  represents the number of advanced swimmers in the club in 2010 and  $b$  represents the number of intermediate swimmers in the club in 2010?

☐ A  $1.53a + 1.49b = 35(1.44)$

☐ B  $1.49a + 0.53b = 35(1.44)$

☐ C  $1.53a + 1.44b = 35(1.49)$

☐ D  $1.44a + 1.53b = 35(1.49)$

18.

$$2l + 2w \leq 27$$

A rectangle has length  $l$  and width  $w$ . The inequality gives the possible values of  $l$  and  $w$  for which the perimeter of this rectangle is less than or equal to 27. Which statement is the best interpretation of  $(l, w) = (8, 3)$  in this context?

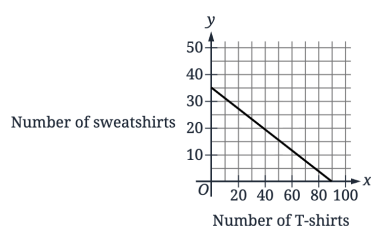
☐ A If the rectangle has length 3 and width 8, its perimeter is less than or equal to 27.

☐ B If the rectangle has length 8 and width 3, its perimeter is less than or equal to 27.

☐ C If the rectangle has length 3 and width 8, its perimeter is greater than or equal to 27.

☐ D If the rectangle has length 8 and width 3, its perimeter is greater than or equal to 27.

19.



The graph models the relationship between the number of T-shirts,  $x$ , and the number of sweatshirts,  $y$ , that Kira can purchase for a school fundraiser. Which equation could represent this relationship?

☐ A  $y = 7x + 18$

☐ B  $7x + 18y = 630$

☐ C  $y = 18x + 7$

☐ D  $18x + 7y = 630$

20.

$$f(x) = 4,000(0.75)^x$$

An entomologist recommended a program to reduce a certain invasive beetle population in an area. The given function estimates this beetle species' population  $x$  years after 2012, where  $x \leq 7$ . Which of the following is the best interpretation of 4,000 in this context?

☐ A The estimated initial beetle population for this species and area in 2012

☐ B The estimated beetle population for this species and area 7 years after 2012

☐ C The estimated percent decrease in the beetle population for this species and area each year after 2012

☐ D The estimated percent decrease in the beetle population for this species and area every 7 years after 2012

21.

At the time of posting a video, a social media channel had 53 subscribers. Each day for five days after the video was posted, the number of subscribers doubled from the number the previous day. Which equation gives the total number of subscribers,  $n$ , to the channel  $d$  days after the video was posted?

☐ A  $n = (53)^d$

☐ B  $n = 53(2)^d$

☐ C  $n = 53\left(\frac{1}{2}\right)^d$

☐ D  $n = (53)^2 + d$

22.

On April 1, there were 233 views of an advertisement posted on a website. Every 2 days after April 1, the number of views of the advertisement had increased by 70% of the number of views 2 days earlier. The function  $f$  gives the predicted number of views  $x$  days after April 1. Which equation defines  $f$ ?

☐ A  $f(x) = 233(0.70)^{\frac{x}{2}}$

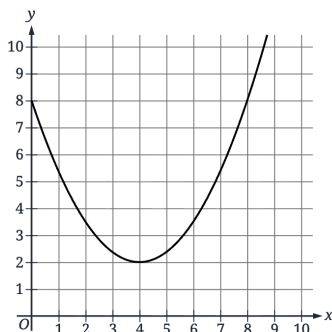
☐ B  $f(x) = 233(0.70)^{2x}$

☐ C  $f(x) = 233(1.70)^{\frac{x}{2}}$

☐ D  $f(x) = 233(1.70)^{2x}$

## Module-2

1.



The graph shows a marble's height above the ground  $y$ , in inches,  $x$  seconds after it started moving on an elevated track of a marble run. Which of the following is the best interpretation of the  $y$ -intercept of the graph?

- ☐ A The marble's height was 0 inches above the ground 8 seconds after it started moving.
- ☐ B The marble's height was 8 inches above the ground when it started moving.
- ☐ C The marble's minimum height was 0 inches above the ground.
- ☐ D The marble's minimum height was 8 inches above the ground.

2.

The function  $f(x) = 240,000(1.22)^x$  gives a company's predicted annual revenue, in dollars,  $x$  years after the company started selling jewelry online, where  $0 < x \leq 10$ . What is the best interpretation of the statement " $f(5)$  is approximately equal to 648,650" in this context?

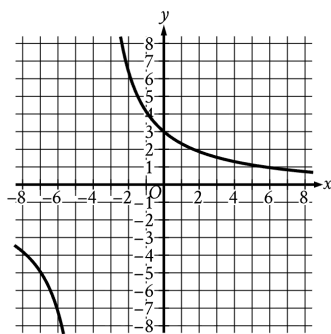
- ☐ A 5 years after the company started selling jewelry online, its predicted annual revenue is approximately 648,650 dollars.
- ☐ B 5 years after the company started selling jewelry online, its predicted annual revenue will have increased by a total of approximately 648,650 dollars.
- ☐ C When the company's predicted annual revenue is approximately 648,650 dollars, it is 5 times the predicted annual revenue for the previous year.
- ☐ D When the company's predicted annual revenue is approximately 648,650 dollars, it is 5% greater than the predicted annual revenue for the previous year.

3.

$$y = -\frac{1}{4}x^2 + 2x + 29$$

The given equation models a company's scheduled deliveries over 8 months, where  $y$  is the estimated number of scheduled deliveries  $x$  months after the end of May 2012, where  $0 \leq x \leq 8$ . Which statement is the best interpretation of the  $y$ -intercept of the graph of this equation in the  $xy$ -plane?

- ☐ A At the end of May 2012, the estimated number of scheduled deliveries was 0.
- ☐ B At the end of May 2012, the estimated number of scheduled deliveries was 29.
- ☐ C At the end of June 2012, the estimated number of scheduled deliveries was 0.
- ☐ D At the end of June 2012, the estimated number of scheduled deliveries was 29.



The graph of  $y = f(x)$  is shown in the  $xy$ -plane. The value of  $f(0)$  is an integer. What is the value of  $f(0)$ ?

5.

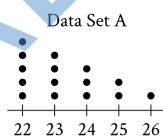
A company designs and makes handbags. To estimate the mean weight of the handbags made by the company on a particular day, a sample of the handbags made by the company on that day was selected at random. Based on the sample, it is estimated that the mean weight of all handbags made by the company on that day is 27.8 ounces (oz), with an associated margin of error of 0.02 oz. Based on this estimate and associated margin of error, which of the following is the most plausible conclusion?

- ☐ A The mean weight of all handbags made by the company on that day is between 27.78 oz and 27.82 oz.
- ☐ B The actual weights of all handbags made by the company on that day are between 27.78 oz and 27.82 oz.
- ☐ C The actual weights of all handbags from the sample are between 27.78 oz and 27.82 oz.
- ☐ D The mean weight of all handbags made by the company on that day is 27.8 oz.

6.

An insurance company offers a series of three information sessions. 1,250 people attended the first information session. 72% of the people who attended the first information session attended the second information session, and 36% of the people who attended the first and second information sessions attended the third information session. How many people attended all three information sessions?

7.



Data set A has 15 values and is represented by the dot plot shown. Data set B is created by adding 46 to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

- ☐ A The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.
- ☐ B The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.
- ☐ C The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.
- ☐ D The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

8.

|                        | 1 visit | 2 or more visits | Total |
|------------------------|---------|------------------|-------|
| Less than 40 years old | 15      | 15               | 30    |
| At least 40 years old  | 20      | 85               | 105   |
| Total                  | 35      | 100              | 135   |

The table summarizes customers who visited a car dealership in the last month by age and number of visits they made to the dealership. If a customer from the last month is selected at random, what is the probability that the selected customer is at least 40 years old?

- A

$\frac{30}{135}$
- B

$\frac{35}{135}$
- C

$\frac{100}{135}$
- D

$\frac{105}{135}$

9.

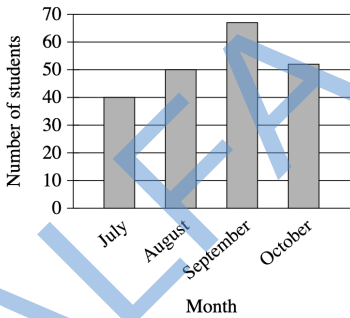
The table summarizes the number of objects in each group.

| Group | Number of objects |
|-------|-------------------|
| A     | 375               |
| B     | 54                |
| C     | 690               |
| D     | 81                |
| Total | 1,200             |

The number of objects in group C is  $p\%$  of the number of objects in group A. What is the value of  $p$ ?

10.

The bar graph shows the distribution of the number of students from one school who were born in one of four months.



How many more students were born in August than were born in July?

- A

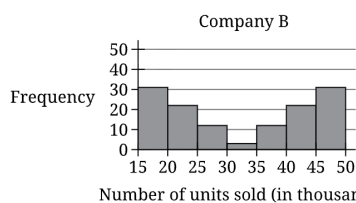
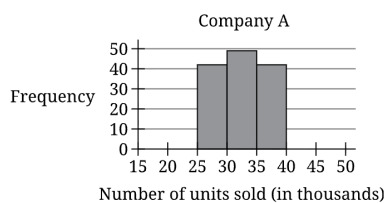
90
- B

50
- C

40
- D

10

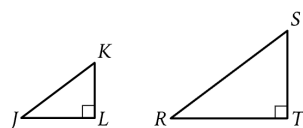
11.



The histograms summarize the distributions of number of units sold, in thousands, for company A and company B. Which statement best compares the standard deviations of number of units sold for these companies?

- ☐ A The standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.
- ☐ B The standard deviation of number of units sold for company A is greater than the standard deviation of number of units sold for company B.
- ☐ C The standard deviation of number of units sold for company A is equal to the standard deviation of number of units sold for company B.
- ☐ D There is not enough information to compare the standard deviations.

12.

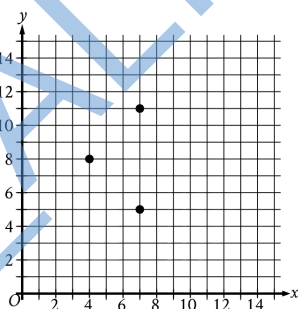


Note: Figure not drawn to scale.

In the figure shown, triangle  $JKL$  is similar to triangle  $RST$ , where  $J$  corresponds to  $R$  and  $K$  corresponds to  $S$ . The length of  $JK$  is 15, and the perimeter of triangle  $JKL$  is 36. The length of  $RS$  is 135. What is the perimeter of triangle  $RST$ ?

- ☐ A 324
- ☐ B 540
- ☐ C 2,916
- ☐ D 8,100

13.



The three points shown define a circle. The circumference of this circle is  $k\pi$ , where  $k$  is a constant. What is the value of  $k$ ?

- ☐ A 3
- ☐ B 6
- ☐ C 7
- ☐ D 9

14.

The table gives the perimeters of similar triangles  $TUV$  and  $XYZ$ , where  $\overline{TU}$  corresponds to  $\overline{XY}$ . The length of  $\overline{TU}$  is 6.

|                | Perimeter |
|----------------|-----------|
| Triangle $TUV$ | 50        |
| Triangle $XYZ$ | 150       |

What is the length of  $\overline{XY}$ ?

- ☐ A 2
- ☐ B 6
- ☐ C 18
- ☐ D 56

15.

In triangle  $DEF$ , the measure of angle  $D$  is  $47^\circ$  and the measure of angle  $E$  is  $97^\circ$ . In triangle  $RST$ , the measure of angle  $R$  is  $47^\circ$  and the measure of angle  $S$  is  $97^\circ$ . Which of the following additional pieces of information is needed to determine whether triangle  $DEF$  is similar to triangle  $RST$ ?

- ☐ A The measure of angle  $F$
- ☐ B The measure of angle  $T$
- ☐ C The measure of angle  $F$  and the measure of angle  $T$
- ☐ D No additional information is needed.

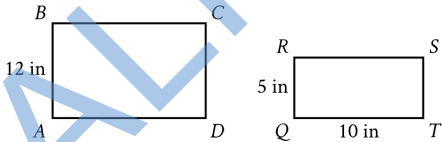
16.

$$x^2 + 58x + y^2 = 0$$

In the  $xy$ -plane, the graph of the given equation is a circle. What are the coordinates  $(x, y)$  of the center of the circle?

- ☐ A (0, 29)
- ☐ B (0, -29)
- ☐ C (29, 0)
- ☐ D (-29, 0)

17.



Note: Figure not drawn to scale.

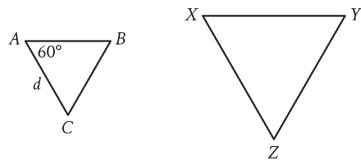
Rectangles  $ABCD$  and  $QRST$  shown are similar, where  $A, B, C$ , and  $D$  correspond to  $Q, R, S$ , and  $T$ , respectively. What is the length, in inches (in), of  $\overline{AD}$ ?

- ☐ A 60
- ☐ B 24
- ☐ C 17
- ☐ D 10

18. The length of each side of a square is 94 centimeters (cm). Which expression gives the area, in  $\text{cm}^2$ , of the square?

- ☐ A  $2 \cdot 94$
- ☐ B  $2 \cdot 94 \cdot 94$
- ☐ C  $4 \cdot 94$
- ☐ D  $94 \cdot 94$

19.



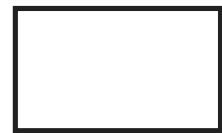
Note: Figures not drawn to scale.

For the triangles shown, triangle  $ABC$  is dilated by a scale factor of 3 to obtain triangle  $XYZ$ , where  $d = 16$ . What is the measure, in degrees, of angle  $X$ ?

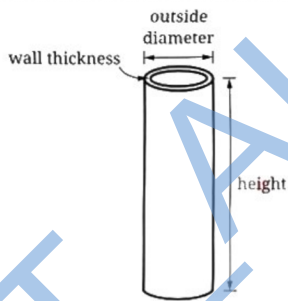
- ☐ A 20
- ☐ B 57
- ☐ C 60
- ☐ D 63

20.

During the first 5.00 seconds after a car started moving, its speed increased to 12.0 meters per second. From 5.00 seconds to 13.0 seconds after the car started moving, its speed increased from 12.0 meters per second to 25.0 meters per second. To the nearest hundredth, what is the positive difference between the average rate of change of the speed of the car, in meters per second per second, during the first 5.00 seconds after the car started moving and the average rate of change of the speed of the car, in meters per second per second, during the period from 5.00 seconds to 13.0 seconds after the car started moving?



21.



Note: Figure not drawn to scale.

In the figure, a hollow steel pipe is in the shape of a right circular cylinder. The steel pipe has an outside diameter of 48 inches, a wall thickness of  $\frac{3}{8}$  inches, and a height of 138 inches. Which of the following is closest to the volume, in cubic inches, of the wall of this steel pipe?

- ☐ A. 61
- ☐ B. 325
- ☐ C. 1,242
- ☐ D. 7,743

22.

To win a game show, a contestant needs to score at least 80 total points from two rounds. Correct responses in the first round are worth 4 points each, and correct responses in the second round are worth 5 points each. Which inequality models this situation, where  $f$  is the number of correct responses in the first round and  $s$  is the number of correct responses in the second round?

A.  $f + s \leq 80$

---

B.  $4f + 5s \geq 80$

---

C.  $5f + s \geq 80$

D.  $5f + 4s \leq 80$